IF ADMIRAL HORATIO NELSON had returned from the grave to view the navies of 1914, he would have been confronted by a startling technological metamorphosis. During a period of only occasional naval warfare, fragile wooden sailing ships with cannon firing broadsides of solid shot had been replaced by warships built of steel, powered by steam, and armed with huge turret-mounted rifled guns firing explosive shells. It was a transformation that had cost vast sums of money, reflecting the importance accorded to navies in a world of competing imperial powers—valued as much for prestige as for use in war. The outbreak of World War I in 1914 belatedly gave steam-and-steel warships a chance to show their paces in a global conflict.



Funnels and masts
Until the 1870s steam warships were built with masts and sails because of concerns about the unreliability of steam engines. These warships of the Argentinian navy are in action on the River Paraguay during the bloody War of the Triple Alliance (1864–70).

NEW SHIPS

The usefulness of a steam engine for making a vessel independent of the wind was obvious in the early 19th century, but the first steam boats were only suitable for rivers, lakes, and coastal waters. Steam gunboats were in use by the 1820s, but it was not until the 1860s that the new face of naval warfare began to emerge. The introduction of the first ocean-going steam warships—although fitted with sails as a supplementary form of propulsion—more or less coincided with the development of naval guns firing explosive shells and the use of iron cladding to protect wooden hulls from this augmented firepower. This revolution in warship design, which threw up numerous strange configurations and combinations, eventually settled down with the Royal Navy's *Devastation* of 1873, a screw-driven steamship with an iron hull, armor, no masts or sails, and guns mounted in turrets instead of broadside. *Devastation* set the style for battleships, which replaced ships of the line as the fleet's capital ships. Frigates were supplanted by cruisers, designed for duties such as convoy escort and commerce raiding. The development of torpedoes led to the introduction of light torpedo boats and also of the torpedo-boat destroyer—soon shortened to "destroyer"—which defended larger ships against torpedo attack and served as a form of torpedo boat itself. When submarines emerged as submersible torpedo boats, the basic outline of a new kind of fleet was complete.



1850 British cannon
In the mid-19th century Royal Navy warships were still armed with muzzle-loading cannon firing solid shot and mounted on wooden carriages. Within 20 years such guns were obsolete.





